

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1711

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 2

Duration : 1 Hour

Total Marks:20

Annexure A

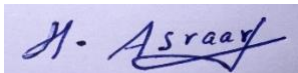
Constraint-Based Problem Solving

Code of Conduct:

I, H ASRAAR AHMED-192524028 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CO1 Assessment: Constraint-Based Problem Solving

1.A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night).

Constraints:

- i. D1 cannot work the Night shift.
- ii. D2 must work before D3 (Morning < Afternoon < Night).
- iii. Only one doctor per shift.
- iv. D3 cannot work the Morning shift.
- v. All doctors must be assigned exactly one shift.

Apply Backtracking Search to find a valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

ANSWER:

Given: A hospital must assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night).

Constraints:

1. D1 cannot work the Night shift.
2. D2 must work before D3.
3. Only one doctor per shift.
4. D3 cannot work the Morning shift.
5. Each doctor must be assigned exactly one shift.

Step 1: Apply Backtracking Search

- Step 1: Assign D1 to Morning. This satisfies the constraint that D1 cannot work Night.
- Step 2: Assign D2 to Afternoon. D2 works before D3.
- Step 3: Assign D3 to Night. D3 does not work Morning.
- Step 4: Check that each shift has only one doctor.

- Step 5: Verify that all doctors have exactly one shift. All constraints are satisfied.

Step 2: Final Schedule

Shift Doctor

Morning D1

Afternoon D2

Night D3

Conclusion: Backtracking Search assigns doctors one by one and checks the constraints at each step. If an assignment violates a constraint, it is undone, and another assignment is tried. The final schedule satisfies all the given constraints.

2. Robot Navigation in a 5×5 Grid (Heuristic Search)

A robot operates in a 5×5 grid environment. It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, right.
- Cost of each move = 1.
- Cannot pass through obstacles.
- Use the Manhattan distance heuristic.

Using the above constraints and the Best-First Search algorithm, show step-by-step node expansion and priority queue updates, and determine the optimal path and its cost.

ANSWER:

Given: A robot must move from Start (S) to Goal (G) in a 5×5 grid while avoiding obstacles.

- Movement allowed: Up, Down, Left, right.
- Cost of each move: 1.
- Obstacles cannot be crossed.
- Heuristic: Manhattan distance.
- Algorithm: Best-First Search.

Note: The assessment PDF does not specify the coordinates of S, G or the obstacles. The following is an illustrative example, not the exact grid from your lecturer.

Step 1: Manhattan Distance Formula

$$h(n) = |x_n - x_G| + |y_n - y_G|$$

Here, $h(n)$ estimates the distance from the current cell to the goal, ignoring obstacles.

Assume Start = (1,1) and Goal = (5,5).

$$h(S) = |1 - 5| + |1 - 5| = 8$$

Step 2: Best-First Search Algorithm

1. Insert the start node S into the priority queue.
2. Calculate the Manhattan distance of its neighbouring cells.
3. Select the node with the smallest heuristic value.
4. Expand the selected node and generate its valid neighbours.
5. Remove obstacles and previously explored nodes.
6. Update the priority queue with the remaining nodes.
7. Repeat until the goal G is reached.

Step 3: Node Expansion

For an illustrative grid, one possible route is:

$$\begin{aligned} (1, 1) &\rightarrow (1, 2) \rightarrow (1, 3) \\ &\rightarrow (1, 4) \rightarrow (1, 5) \\ &\rightarrow (2, 5) \rightarrow (3, 5) \\ &\rightarrow (4, 5) \rightarrow (5, 5) \end{aligned}$$

Step 4: Priority Queue Updates

Step	Expanded node	Next selected node	Heuristic
1	(1,1)	(1,2)	7
2	(1,2)	(1,3)	6
3	(1,3)	(1,4)	5
4	(1,4)	(1,5)	4
5	(1,5)	(2,5)	3
6	(2,5)	(3,5)	2
7	(3,5)	(4,5)	1

8	(4,5)	(5,5)	0
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The search selects the node with the smallest heuristic value at each step. Other valid neighbouring nodes may also remain in the priority queue.

Step 5: Calculate Path Cost

The example path contains 8 moves, and each move costs 1.

$$\text{Total Cost} = 8 \times 1 = \boxed{8}$$

Conclusion: Best-First Search uses the Manhattan distance heuristic to guide the robot toward the goal while avoiding obstacles. The example path costs 8. However, Greedy Best-First Search does not guarantee an optimal path in every grid.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided